

Name: _____

Date: _____

RATIO, (DIVIDING IN A RATIO)

LO: I can divide a quantity in a given ratio.

Dividing in a ratio

Dividing in a ratio means splitting a total into parts according to a given ratio. Find the total number of parts first.

Method: Add the ratio parts. Divide the total by this sum to find one part. Multiply to find each share.

Quick examples

●	£20 in ratio 3:2 = £12 and £8	5 parts, £4 each
●	Ratio 1:3 means one quarter and three quarters	fractions of total
●	Total parts = sum of ratio numbers	first step
●	£20 in ratio 3:2 = £10 and £10	divided equally, not in ratio

Key idea

Add ratio parts. Divide total by number of parts. Multiply each ratio number by value of one part.

$$\text{Total} \div (a + b) = 1 \text{ part}$$

Then multiply: a parts and b parts

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – simple ratio

Divide £35 in the ratio 3:4.

$$\text{Total parts: } 3 + 4 = 7$$

$$\text{One part: } 35 \div 7 = £5$$

$$3 \text{ parts} = £15, 4 \text{ parts} = £20$$

£15 and £20

$$\text{Check: } £15 + £20 = £35.$$

Example 2 – three-part ratio

Divide 60 kg in the ratio 1:2:3.

$$\text{Total parts: } 1 + 2 + 3 = 6$$

$$\text{One part: } 60 \div 6 = 10 \text{ kg}$$

$$\text{Shares: } 10 \text{ kg, } 20 \text{ kg, } 30 \text{ kg}$$

10 kg, 20 kg, 30 kg

Same method works for three or more parts.

Example 3 – finding the total

Money is shared in ratio 5:3. The larger share is £40. Find the total.

$$\text{Larger share} = 5 \text{ parts} = £40$$

$$\text{One part} = 40 \div 5 = £8$$

$$\text{Total} = 8 \times 8 = £64$$

Total = £64

Work backwards: find one part from the known share.

Example 4 – finding the difference

Divide 80 in the ratio 7:3. Find the difference between the shares.

$$\text{Total parts: } 10. \text{ One part} = 80 \div 10 = 8$$

$$\text{Shares: } 56 \text{ and } 24$$

$$\text{Difference: } 56 - 24 = 32$$

Difference = 32

$$\text{Or: difference} = (7-3) \times 8 = 4 \times 8 = 32.$$

YOUR TURN – PRACTICE (deliberate practice)

1. Simple two-part ratios

Divide each amount in the given ratio.

- a) £24 in ratio 1:3
- b) £40 in ratio 3:5
- c) 60 cm in ratio 2:3
- d) £100 in ratio 7:3

2. Three-part ratios

Divide in the ratio given.

- a) £60 in ratio 1:2:3
- b) 120 kg in ratio 2:3:5
- c) £180 in ratio 4:5:1
- d) 90 litres in ratio 1:2:6

3. Finding the total or one share

Work backwards from the given information.

- a) Ratio 2:5. Smaller share = £14. Total?
- b) Ratio 4:3. Larger share = £48. Total?
- c) Ratio 5:2. Difference = 21. Total?
- d) Ratio 3:7. Smaller share = 12 kg. Larger share?

4. Word problems

Solve each ratio problem.

- a) 3 friends share £45 in ratio 2:3:4. How much does each get?
- b) Concrete: cement:sand:gravel = 1:2:4. Need 35 kg total. How much sand?
- c) Ages in ratio 3:5. Older is 20 years. How old is younger?
- d) Profit shared 2:3. Total profit £7500. How much more does larger share get?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To divide in ratio 3:2, find total parts (5) first ☐

Correct answer: _____

Reason: _____

b) Ratio 3:2 means divide by 3 and by 2 ☐

Correct answer: _____

Reason: _____

c) The shares should always add up to the original total ☐

Correct answer: _____

Reason: _____

d) £20 in ratio 3:2 gives £10 each ☐

Correct answer: _____

Reason: _____

e) Ratio 1:4 means one fifth and four fifths ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Three siblings share an inheritance of **£54000** in the ratio of their ages: **18, 24, and 30**. Simplify the ratio first. How much does each sibling receive?

Expression: _____

Simplified: _____

b) A drink is made from juice, water, and fizz in the ratio **2:5:3**. If you have **400ml of juice**, how much drink can you make in total? How much water and fizz do you need?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

